

DEMYSTIFYING ERGATIVITY

Hubert Haider

Dept. of Linguistics, Univ. Salzburg

Abstract

Facts fall in place and form a coherent picture once myths have been dispelled. Descriptions of ergativity are fraught with grammatical myths such as: No ergative SVO languages; optional omission of subjects; objects agree, even if subjects don't; grammatical voice eliminates objects and the subjects switch into the object case; subjects are syntactically less accessible than objects, and so on. Strikingly, all the puzzlement stems from a single source and dissolves into nothingness as soon as one is willing to acknowledge the origin of these mystifications, namely the persistent misinterpretation of the grammatical-subject function of ergative languages, by functionalist typologists and by generativist grammarians. Ergative languages and nom-acc languages follow the same principles, *but* modulo inverse alignment of the subjects.

The article is organised as follows: Section 1 describes the current predicament. Section 2 presents the central problem and the principal way out of the self-inflicted dilemma, which is a consequence of the currently widely held assumption that ergative is the subject case. Section 3 argues step by step that absolutive rather than ergative is the subject case. Data and arguments in the literature that purport to prove ergative as the subject case do not stand up to close scrutiny. Finally, Section 4 dispels the OVS myth. In this survey, a central prediction is tested and it shows that virtually all so-called OSV languages are ergative languages and thus SVO.

1. Background. It is high time to break with the antiquated *communis opinio* that the ergative is the subject case.¹ Typologists have become accustomed to viewing ergative languages as languages with grammatically 'exotic' subject behaviour. It is time for an empirically more successful theory that does not have to contend with a multitude of undesirable side effects of the currently prevailing view. It is sufficient to realise that one has made a mistake in identifying the subject.

In short, ergative is *not* the case of the grammatical subject and absolutive is not the case of the *object* of a transitive verb. The absolutive is the case that marks *intransitive* subjects and therefore, it is the subject case of an abs-erg system. Ergative is the *dependent case* that marks the 'other' core argument of transitive verbs with structurally determined cases (see Figure 1 below). What matters grammatically is the parallel *syntactic* behaviour of nominative and absolutive. The fact that nominative and ergative are semantically linked to the very same core argument of the lexical argument structure of verbs (viz. same semantic role) is totally irrelevant for the grammatical constitution of the respective subject relations.

An *intransitive finite sentence* is the minimal setting of a subject-verb configuration in any language whose grammar defines a grammatical subject. Consequently, this configuration is the model for the subject case in a given language. Thus, the case of the intransitive subject is also the *subject case* of a clause with more than one argument, and in particular, with a transitive verb in a subject-object configuration.

¹ "Ergative: case inflection marking transitive subject (A)". [glossary of Dixon (2020 vol. 2)].

At this point, already the first myth disappears: Ergative languages with an Abs-V-Erg order are SVO languages and not, as hitherto still maintained, OVS languages (see section 4.). [O[VS]] languages – as Joseph Greenberg had foreseen from the outset – do not exist, neither as ergative languages nor as accusative ones.

The currently prevailing view is needlessly biased. Its maxim is simple, easy to apply,² and false. According to this maxim, the respective core argument of a verb that is the grammatical subject in a nominative-accusative language, such as Latin³ or English, is also the grammatical subject in corresponding sentences of every other natural language. In other words, subjects would translate into subjects in any language. This may sound plausible to some linguists, but it is by no means so. Data and reasons will follow. Here are a few particularly embarrassing consequences that are triggered by assuming – contrary to the facts – that in an ergative language, the ergative-marked noun or phrase is the *grammatical* subject of a finite sentence

- **No SVO?** Although the SVO-type is one of the major word order types, there are no ergative “SVO” languages (viz. Erg-V-Abs), but numerous languages are attested as ergative “OVS” (that is: Abs-V-Erg), a virtually inexistent word order type under nom-acc alignment. A nom-acc [Acc [V Nom]] language has never been detected.
- **Optional subjects?** In a considerable number of languages, the ergatively marked core argument is grammatically optional and can be omitted, but the absolutive is obligatory. If the ergative is regarded as the subject, this is the exact opposite of mandatory nominative subjects but optional accusative core arguments in nom-acc languages. If this were an option, it would be expected to be seen also in nom-acc language, which is not the case.
- **Subject-to-object?** In relation-changing diatheses (e.g. antipassive), ergative switches into the absolutive, which would amount to a subject case being demoted to an object case whenever the original object is eliminated. In comparison, no nom-acc language shows such a behaviour, that is, a nominative switching into an accusative whenever the original direct object is omitted. However, accusative changes into nominative whenever the subject is omitted, and so does the ergative.
- **Not accessible?** In languages with accessibility restrictions in relative clause formation, the ergative (as the alleged subject) may be less accessible than the absolutive (as an alleged grammatical object). Here, too, there is no parallel in nom-acc languages.
- **Only finite?** Ergative noun phrases can occur in non-finite contexts (*clausal* infinitival constructions without agreement morphology) in which a subject in the form in which it occurs in a finite sentence ought to be excluded.

What the “*Agent=Subject*” maxim completely neglects is the fact that in the languages of the world, there are alternative grammatical ways of alignment, that is, ways of linking the core arguments of verbs (and other argument-taking lexemes) to their syntactical representatives (see Figure 1). In languages with ergative alignment, unlike in nom-acc, the candidate for the

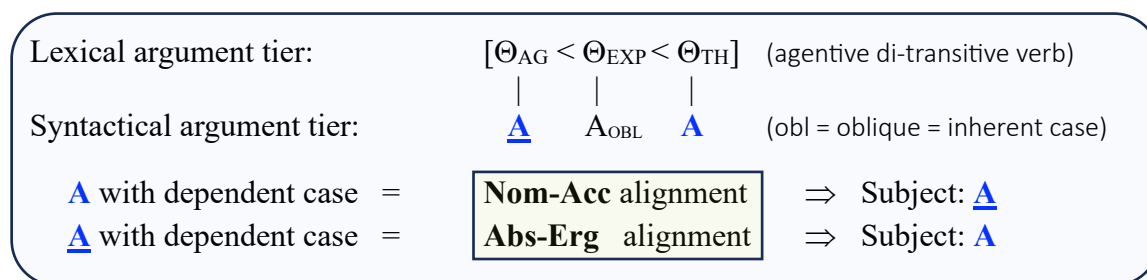
² Queixalós and Gildea (2010: 8) are explicit in this respect: “*So for now we adopt the theoretically problematic but heuristically useful practice of relying on intuitive-impressionistic identifications of A and P.*” On the descriptive level, this is fully legitimate. It is only the next step that is fatal, namely equating A with “subject.”

³ During the 20th century, English has begun to replace Latin as model grammar. In Latin, subject is defined in terms of case and agreement. In the grammar of English, subject is a configurational notion. Virtually all current grammar models have been designed by native speakers of English, whence their particular biases.

grammatical subject is not the *highest*-ranked⁴ core argument of the lexical argument structure of a verb with structural⁵ case, but the lowest-ranked one. In brief, the abs-erg linking relation appears to be the inverse of nom-acc linking.⁶

The rest follows from the very same, independently motivated principles. The principles are identical across alignment systems but the outcomes are different because of the basic difference in aligning the argument structure to the morpho-syntactical structure. In short, this approach results in a simple theory: the same principles, but predictably different results due to the *reverse linking* in the alignment; see Figure 1.

Figure 1: Alignment parameter (Haider 1985a,b, 1986; Van Valin 1992).



Within Generative Grammar, the – in fact self-evident – idea that the absolutive is the subject case of ergative languages was proposed already in the eighties by Marantz (1981, 1984) and developed further by Levin (1983); see Aldridge (2008). The reason why it failed to gain acceptance was that it was based on implausible, theory-internal assumptions, without convincing independent evidence. In their proposal, the entire theta-assignment contrivance of ergative systems is seen as an ‘downside-up’ version of the nom-acc mechanism: As for nom-acc systems, the theta role of the subject was – and still is – deemed to be assigned by the whole VP (= predicate) or, even more mysteriously⁷, by a silent v° , while all other roles are assigned by the verbal head to VP-internal positions. In short, in nom-acc, the *agent role* is assigned VP-externally. And here comes the downside-up, ergative version: In abs-erg, the *theme role* is taken to be assigned by the whole VP while the agent is part of VP-internally V° -assigned roles. Not surprisingly, no independent evidence came forward to support such a creative claim.

This idea was obviously too ad hoc to find any long-lasting support. Looking back, it's clear that Marantz and Levin barked up the wrong tree. What is important and causal is not the details of theta assignment, but rather the simple parametric selection of the candidate for the subject case in the lexical argument structure of a transitive verb. Everything else works exactly the same way in both systems, especially with respect to the case-dependency management.

Let me add in passing that this article is *not* an article about a terminological issue. It is about a question of content. Evidently, whenever people confound subjects with objects, this must

⁴ The arguments of a verb are ranked in the argument structure. The ranking reflects the positions in the hierarchically organized lexical argument structure: [AGENT [EXPERIENCER/RECIPIENT [THEME]]]. See e.g. Jackendoff (1990).

⁵ Unlike *inherent* aka *lexical* case, which is specified in the lexical argument structure, the case of noun (phrases) with *structural* case is not prespecified but determined by their syntactic environment.

⁶ This has been noted in Haider (1986), with accusative and ergative as the respective dependent cases.

⁷ This is an entirely theory-internal hypothesis without compelling empirical basis (see Haider 2020). Here is an immediate counterexample: “It literally and figuratively *rained coins*.” Evidently, the semantically empty “it” is an argument of the weather-verb, and neither of a whole VP nor of a “little v° ”: #“We rained cats and dogs.”

have unpleasant empirical consequences. Here is a telling example. It concerns a seemingly enigmatic effect of the alignment split on the identification of the antecedent in coordination reduction constructions in Yidin^y.

Dixon (1994: 177) concludes his discussion of the grammatical pivot of coordination in Yidin^y with a sigh: “*It would be nice if one could uncover a universal rationale for the occurrence of S/A and S/O pivots in a single language.*” You’re welcome! The key concept (see sect. 3) is “grammatically privileged core argument” (= grammatical *subject*). If a language has a split alignment system with aligning *non-pronominal* core arguments by abs-erg and *pronominal* ones by nom-acc, this results in two different contexts of qualifying for a grammatical subject.

Yidin^y is such a language, see Dixon (1994: 176). First, “*nouns inflect on an ergative pattern and pronouns on an accusative one (1st and 2nd person)*”. Second, “*Yidin^y has an S/O pivot for all types of relativisations and one kind of coordination, but an S/A pivot for another kind of coordination.*” The “*another kind of coordination*” is one with a pronominal antecedent: “*There is an S/A pivot for joining together two clauses which have a common NP that is pronominal, and an S/O pivot for linking clauses whose common NP is non-pronominal.*” Pronominal means personal pronouns. So, the *grammatical subject* is S (= Nom) and A (= Nom) for pronominals, but S (= Abs) and “O” (= Abs) for non-pronominal arguments, and the pivot is in each case grammatically identical, namely the *grammatical subject* (with absolutive and nominative as subject cases in alternative contexts).

This is an on-target showcase of the explanatory power of an empirically adequate and precise concept of “grammatical subject”. Dixon’s pivot of coordination-reduction is the *grammatical subject*. A split alignment system (viz. abs-erg & nom-acc) has two different contexts for grammatical subjects, namely one in an absolutive-ergative mode and another one in a nominative-accusative mode. Consequently, such a language has more than one type of pivots, either the absolutive in abs-erg contexts or the nominative in pronominal nom-acc contexts, in the very same language.⁸ Yidin^y has no person/number marking on the verb. It is dependent-marking. The grammatical functions such as subject and object are marked on the noun (phrases) only, that is, by flagging the NPs, not by indexing the finite verb.

If a language has *both*, flagging (viz. case) and indexing (viz. agreement on the verb), the two systems go usually parallel, but splits are possible, as for instance in Pesh/Pech/Paya, in the Carbón variety (Chamoreau 2021), with abs-erg alignment for relational markers (flagging) and a nom-acc pattern for agreement (indexing), which treats transitive subjects and intransitive ones alike. What is the grammatical subject in such a case? The answer is straightforward. Subject is the privileged argument. In Pesh, the privilege manifests itself in congruence, not in case. There is no implementation of dependent case, no passive or antipassive, and – as predictable therefore – the typical subject-controlled properties (e.g. conjunction reduction, control) are nom-acc-typical congruence patterns, since Pesh ‘privileges’ the agentive argument of transitives, treating it on a par with the single argument of intransitives.

⁸ Finding out how the elegance of the grammar of Yidin^y becomes evident under the right perspective is a pleasure. In order to see this, one has to only realise that an absolutive is a *subject* case on a par with nominative. Even a man whom I greatly respect as a scholar with a keen sense for the foundations of grammar – R.M.W. Dixon – was distracted by the mischaracterisation of the subject function from making out the otherwise evident connection.

An equally revealing case comes from Warlpiri (Dixon 1997:172-174). In coordinations, the omitted argument is identified in a nom-acc mode although the language is consistently case-marked in the ergative mode. The key is here: “*Case marking on NPs is ergative, but cross-referencing suffixes in the auxiliary are quite accusative*”. The parallelism restriction on coordination reduction requires that the omitted element in the second conjunct be grammatically parallel to its antecedent. Since an omitted element cannot be checked for overt case, the agreement system takes over, and this works in the nom-acc mode. Therefore, we see a pivot typical of nom-acc in Warlpiri, even though it is a (split) abs-erg language.

2. Ending a mistold story – Five generalisations and their implications

The still prevailing opinion on ergativity is described in an online encyclopaedia⁹ as follows: “*Ergative-absolutive languages, sometimes called ergative languages*” are “*languages where the subject of an intransitive verb and the object of a transitive verb behave the same way in a sentence.*” Evidently, this description is worded in the categories of languages with nom-acc alignment. The argument of the verb that is “*the object of a transitive verb*” is an *object* only in a nominative-accusative setting. In an absolutive-ergative system it is the *subject* of the clause.

An unbiased and empirically adequate rendering would be this: In the vast majority of languages, one of the two arguments of a transitive verb, X and Y, is aligned with the syntactic subject function and the other one with the syntactical object function. Consequently, this opens a system potential with at least two options. In one, argument X is linked with the subject function, in the other, argument Y is the subject. In each system, the remaining argument is linked to the grammatical-object function. If X is the *agent* argument of an agentive verb linked as subject, the alignment system is called *nom-acc*. If Y is the *non-agent* argument of an agentive verb linked as subject, the alignment is called *abs-erg*.¹⁰

That the ergative-marked noun or noun phrase is regarded as the grammatical subject in an ergative system is a misnomer with a long history – e.g. Uslar (1889:122-123) or Schmidt (1902) – rooted in a centuries-old grammar tradition, but the underlying theory and its outcome are inelegant and not convincing enough to have a realistic chance to be true. A straightforward perspective based on current knowledge considers a nominal expression to be a grammatical subject if it behaves as unbiased syntacticians predict a subject to pattern, from a *grammatical* point of view. “Subject” is a formal concept based on formal properties. It cannot be adequately defined solely by reference to its typical semantic content in a nom-acc language, viz. “agentive”, as the old school prefers to do.

The following five generalizations (G.1 - G.5) are based on grammatical core properties. They collectively reflect the syntactical equivalence between nominative-accusative and absolutive-ergative systems at the level of grammatical relations, namely subject vs. object, despite their surface differences in case marking patterns, which immediately follow from a clear-cut alignment parameter.

G.1: In a prototypical *nom-acc* language, the grammatical subject of a finite clause with a *single* nominal (non-pronominal) core argument is a *nominative* noun phrase (Status:

⁹ Wikipedia on “*Ergative-absolutive language*”.

¹⁰ There is even room for instantiating either option, viz. one for full noun phrases and one for pronouns. This is known as a split ergative system.

uncontroversial).

G.2: In a prototypical *abs-erg* language, the grammatical subject of a finite clause with a *single* nominal (non-pronominal) core argument is an *absolute* noun phrase. (Status: uncontroversial).

G.3: In finite clauses, arguments with dependent case can be omitted but a (non-pronominal) subject argument cannot. (Status: uncontroversial).

Comment: Here is an English example. The verb ‘forgive’ can be used with the subject only, or with subject plus direct or indirect object, or with all three arguments, but it cannot be used without the subject argument, unless passivised or in a middle construction. In ergative languages, the argument with absolute case must not be omitted, but the ergatively marked noun phrase (often) can. [Dixon (2010: 168)].

G.4: If the default argument for the subject function of a finite clause remains implicit, this is signalled morpho-syntactically and/or structurally. (Status: uncontroversial).

Comment: In nom-acc languages, this is traditionally called passive. English, just like many other Indo-European languages, employs a verb form (= participle) that blocks the argument that would surface as subject and combines this form with an auxiliary that does not undo the blocking (viz. *be* vs. *have*). In other languages, as for instance Japanese, Latin, or Turkish, the blocking of the subject argument is signalled by a ‘passivising’ affix on the verb.

In ergative languages, the core argument whose omission is signalled morpho-syntactically is not the ergative; it is the noun phrase with *absolute* case. This phenomenon is traditionally called antipassive. In fact, it is nothing else but the passive construction in the setting of an ergative language:¹¹ The argument that would be represented by the subject is omitted, which is morpho-syntactically signalled, and the argument with the dependent case (= ergative) replaces it as grammatical subject and surfaces in the subject case, that is, the case that the eliminated argument would have received (see G.5), namely absolute.

G.5: In a language with structural cases, the case of the subject of a finite intransitive verb is also the case of the subject of a transitive verb. If the default subject argument is eliminated, the candidate for the dependent case is assigned the subject case. (Status: to become uncontroversial).

Comment: The causal moment is a dependency¹² relation between subject and object cases in a finite clause: No structural object case in the absence of a structural subject case. The dependency reflects a “push-down-stack” relationship (cf. the so-called object-to-subject ‘promotion’), where ergative switches to absolute whenever the candidate argument for the subject case is absent. The parallel is evident: Nominative is the default

¹¹ “Ergative systems have an analogous construction, here termed as antipassive, which has all the properties of the passive, as Kuryłowicz, again saw.” Silverstein (1976: 140), who is the namesake of ‘antipassive’.

¹² Acc-assignment is dependent on nom-assignment [Haider (1986: 121)]. In ergative languages, the ergative case is the dependent case [Haider (1986: 122)]. So, ergative assignment depends on abs-assignment. Antipassive does in the abs-erg setting what passive does in the nom-acc setting. Passive and antipassive are grammatical equivalents, modulo alignment type. [Haider (1986: 123)].

case in nom-acc languages: An accusative as the case of the direct object switches to nominative in the absence of the subject. In abs-erg languages, absolutive is the default case. Ergative switches to absolutive. What is different merely is the mode of alignment.

The five generalisations are premises of the following six conclusions:

- C.1: In nom-acc languages, the subject case is *nominative*; in ergative languages it is *absolutive*. (G1, G2, G5).
- C.2: The *nominative* subject of a finite nom-acc clause shares syntactic core properties with the *absolutive* subject of a finite abs-erg clause. (G4-G6).
- C.3: Canonical passive in nom-acc and canonical antipassive in abs-erg are *one and the same* general grammatical device, modulo alignment type. (G4, G5).
- C.4: If the primary subject is morpho-syntactically eliminated, the argument with a *dependent* case is assigned the case of the subject (G5). So, the accusative case of the direct object is replaced by the nominative in passive (G1), and the ergative case is replaced by the absolutive in antipassive (G2). In each instance, a dependent case is replaced by the subject case.
- C.5: In a nom-acc language, the *accusative* argument can be omitted with many verbs; in an abs-erg language, the *ergative* argument can be omitted with many verbs. In each case, the omitted argument is treated as an implicit argument and semantically interpreted by means of existential closure.¹³ (G1, G2, G3).
- C.6: The subject of an intransitive finite verb has the same grammatical form (viz. case or structural setting) as the subject of a finite transitive verb. Nominative and absolutive are subject cases. The reason is as follows: Objects can be omitted, subjects cannot (G3). In nom-acc languages, the nom-argument is obligatory; in abs-erg languages it is the absolutive. The ergative can be omitted. This turns a transitive construction into an intransitive one, or more precisely, in an unaccusative or unergative one, respectively.

Next, let us compare the claims just outlined with the currently dominant view. It is hard to overlook the fact that the old-school view is a clumsy narrative of striking inelegance,¹⁴ and on top of it, that it contradicts relevant empirical findings. In this ‘theory’, that could just as well be called grammatical mythology, the grammatical subject of a sentence is identified based on the *semantics* of one of the core arguments of a transitive verb.

The text on the cover of Dixon (1984) is a locus classicus: “*The term 'ergativity' is used to describe a grammatical pattern in which the subject of an intransitive clause is treated in the same way as the object of a transitive clause, and differently from transitive subject.*” In Dixon (2010, vol. 2: 119), he chose the following formulation. Cross-linguistically, “*there are two recurrent patterns – S marked like A and S marked like O.*” This sounds like a practical rule of thumb, but empirically, it is misleading. Why would the intransitive subject share the case of an object in an ergative language? In other words, the subject of a transitive verb is considered

¹³ Existential closure is a formal device that introduces an existential quantifier to bind free variables. Example:

i. They hunt and gather $\Rightarrow \exists x \exists y$ [they hunt (x)] & [they gather (y)]. (There is something they hunt/gather)

¹⁴ “*Elegance is a prized quality in science that is associated with simplicity and explanatory power.*” Casadevall & Fang (2018).

to be the noun phrase that represents the agent(-like) argument, under any type of alignment, no matter what the cost.

In the literature on ergativity, there is no plausible explanation as to why the subject of an intransitive sentence, the prototypical subject of a finite sentence, should be “disguised” with the object case of a transitive verb, namely the absolutive.¹⁵ The strange thing about this is not so much the resulting odd grammatical picture, but rather the way in which clear facts are misperceived.

The appropriate formulation, as argued in this article, requires only a rearrangement of Dixon’s own words: *There are two recurrent patterns – On the one hand, A is marked like S and on the other hand, P is marked like S*, (with A and P denoting the two structurally case marked core arguments of a transitive verb). The first pattern is the nom-acc setting, the second setting the abs-erg one. In other words, in nom-acc languages, the A-argument of a transitive verb is the subject, while in ergative languages, the *other* core-argument of a transitive verb is the subject of a clause with a transitive verb. In each case, “S”, that is the subject of a monovalent verb, defines the grammatical reference point for “subject”. A noun phrase that morpho-syntactically behaves like S is the subject in any alignment setting, nom-acc or abs-erg.

It is an irony of linguistic history that one of the few assumptions shared by the school(s) of functional typology and the grammar creations in the wake of Noam Chomsky is precisely the assumption that the grammatical subject of a transitive verb is generally the representative of the argument that designates the agent. Generative grammarians of the Minimalist Program era – see e.g. Lahne (2008), Roberts (2021), Taraldsen (2017) – share the typologists’ preference of a universal *Theta-role-based* characterization of the *syntactic* category ‘subject’ and are inevitably confronted with the logical truth of “ex falso quodlibet” (from falsehood, anything follows).¹⁶

A theoretical underpinning of the “agent = subject” viewpoint is derived from an axiom by Baker (1988, 1997), namely the “Uniformity-of-Theta-Assignment Hypothesis”. It states that *identical thematic relationships* between predicates and their arguments are represented by *identical structural relationships* between these items in the syntactic base structure. This may be largely the case *within* a particular language,¹⁷ but definitely not across alignment systems, since it neglects the essential moment, namely the grammatically determined option for the parametrical choice function for an argument to serve as grammatical *subject*.

The prevailing view – the representative of the agent role is the default syntactical subject – seems to be so firmly entrenched that people are willing to steadfastly disregard how badly it

¹⁵ In his Grammar of Warrongo, Tsunoda (2011) shows himself to be an uncompromisingly consistent grammarian, albeit in the opposite direction. The absolutive is renamed as accusative and so, an ergative language would be an erg-acc language. This makes the grammatical inconsistencies even more perspicuous.

¹⁶ This insight of mediaeval logicians (Guillaume de Soissons, Paris, 12th cent.) highlights that inconsistent systems lack the ability to distinguish between truth and falsehood.

¹⁷ But even within the same language (e.g. German), there are non-matching doublets. (i) means the same as (ii), and (iii) means the same as (iv). Such doublets are unexpected under the putative regime of the axiom:

i. dass sie _{Nom} den Vater _{Acc} entbehrt hat that she the father lacked has ‘that she missed her father’	ii. dass ihr _{Dat} der Vater _{Nom} gefehlt hat that her the father lacked has ‘that she missed her father’
iii. dass er _{Nom} mehr als drei Häuser _{Acc} besitzt that he more than three houses owns	iv. dass ihm _{Dat} mehr als drei Häuser _{Nom} gehören that (to)him more than three houses belong

fits with data from ergative languages in a cross-linguistic perspective on grammar theory. Although the absolutive noun phrase exhibits core properties of a syntactic subject, the bias rooted in the study of Indo-European languages was stronger; see also Gil (2001) on Eurocentrism. In any Indo-European language, the grammatical subject of a typical transitive verb is the noun phrase that represents the agentive core argument.¹⁸ ‘Therefore’ (= bias), this quality tells what qualifies as a subject universally. In the nom-acc systems of Indo-European languages this is correct, of course, but as a universally valid generalisation, it is completely off the mark.

Grammatical subject is a notion grounded in *grammatical* properties. Simply put, the subject is the *grammatically privileged* core argument of a simple clause, as suggested by Van Valin (1992) or by Mel’čuk (2014: 179).¹⁹ The ways in which the privileges manifest themselves depend on the particular morphosyntactic and structural inventories of the respective languages. Across languages, “grammatically most privileged” is a relative quality.²⁰

Subject-verb agreement, for instance, is not a universal property of subjects, but if a language has an agreement system with argument-verb agreement (e.g. number or person), subjects agree with the finite verb. Conversely, there are no languages in which core arguments obligatorily show number & person agreement with the verb but subjects don’t. The traditional theory of ergativity therefore is challenged by ergative languages with *absolutive*-verb agreement in the absence of verb-*ergative* agreement. In the framework of the traditional view, this unmistakably amounts to unexpected cases of object agreement in the absence of subject agreement (cf. Woolford 2000; Coon 2017).

Case-dependency is another example (Haider 1986, 2000; Marantz 1991, Baker 2015). This means that the licensing/assignment of *default* structural case has priority over the assignment of *dependent* structural case. In other words, the dependent case can be assigned only in the presence of the default case. The dependency relation is a push-down-stack relation. An immediate consequence is the switch from dependent case to subject case whenever the candidate for subject case (= default case) is missing. In this instance, an argument with dependent structural case becomes the highest argument in the stack and appears as if ‘promoted’ to the case of the subject. Every linguist knows this. If, for instance, the primary subject candidate has been grammatically neutralised, as in the passive/anti-passive constructions, there is a switch from ergative to absolutive. A corresponding switch is acc-to-nom, in acc-nom languages. In each case, the grammatical trigger is the very same dependency relation.

The reverse ‘switch’, as implicated by the old-school view on ergative-to-absolutive, does not exist. There is no nom-acc language in which the *omission* of an object would cause the nominative to change into an accusative.²¹ By the same token, there is no ergative language in which the absolutive switches into an ergative, when the ergative candidate is omitted, although this

¹⁸ The Indo-Aryan ergativity is a grammatical accident in the change of perfect-tense formation (see Section 5).

¹⁹ I would like to thank Martin Haspelmath for pointing out Igor Mel’čuk’s contribution to me.

²⁰ Role & Reference Grammar (Van Valin 1992), as presented in Van Valin (2005: 99), entirely avoids the concept of grammatical subject since “*subject and direct object are not universal and cannot be taken as the basis for adequate grammatical theories.*” What he means is “subject” in the traditional usage. But, would a physicist reject the concept of the atom because there is no *universal* atom?

²¹ This happens only when a superordinate nominative argument is introduced, as an effect of *causativisation*, for instance: *He laughs – This made him laugh*. The exact ergative counterpart is the *applicative* construction: An additional subject argument is introduced as absolutive and the original absolutive subject switches into the dependent case, viz. ergative.

would be an unspectacular object-to-subject switch. However, in the traditional view, people seem to be willing to accept that in antipassives, a ‘subject’ (ergative) ends up as in the case of an ‘object’ (absolutive). Shouldn’t such a phenomenon be considered just as extremely surprising and improbable as a nominative case changing to an accusative case whenever the object of a transitive verb is omitted? Any reasonably knowledgeable syntactician must be appalled by such assumptions and predictions, or rather mystifications.

Conclusion C.4 – Keenan’s (1976) indispensability property of syntactic subjects – applies to nom-acc and abs-erg languages in the very same way. Grammatical objects may be omitted (under certain restrictions), subject arguments cannot be omitted freely. This is what the empirical facts tell us. The ergative core argument may be instantiated optionally, just like the accusative object with many verbs. Ergative “subjects” are “dispensable”,²² in Keenan’s sense, as described, for example, by Churchward (1953: 68) for Tongan, by Keenan (1976: 313) for Tongan and Tibetan, by Fernández & Berro (2022) for Basque, by Haig (2008, 2017) for Kurmanjî Kurdish (“agentless perfective”) or by Dixon for Dyirbal:

Dixon (2010:168) is a rare testimony, since he also pays attention to the absolutive: “*In an ergative language like Dyirbal, S and O arguments are obligatory and A optional.*” This means, the ergative is optional, the absolutive mandatory and this is a well-known subject-object asymmetry, with absolutive and nominative being the cases of subjects, respectively. In sum, the absolutive cannot be optionally omitted in the same way as the ergative can be in ergative languages

On the other hand, the omission of nominative and absolutive subjects must be morpho-syntactically signalled (by passive/antipassive markers) while an accusative object or an ergative-marked core argument can *ceteris paribus* be omitted without further ado. This, too, should have raised serious doubts as to whether an ergative noun phrase could be reasonably regarded as a grammatical subject.

Given such circumstances, one has to ask why anyone would cling to the idea that the ergative case is the case of the grammatical subject.²³ The only²⁴ plausible reason for this, in my opinion, is the misjudgement of ergative languages as highly exotic compared to Indo-European languages. Only from this perspective²⁵ could one maintain the biased notion that the grammatical subject of a typical transitive verb is universally associated with the role of the agent. Ergative would be the exotic subject case and the absolutive has to be downplayed to an exotic object case since one cannot deny that it is the case of the subject of intransitive finite verbs.

²² “Dispensable” means that a core argument may be omitted. Semantically, the argument is then interpreted by existential closure, that is, as indefinite and unspecific. “*He won*” means that there is *something* that he won. The dropping of dispensable arguments (ergative in abs-erg, and accusative in nom-acc) must not be confused with the dropping of pronominal arguments (“pro-drop”). They are interpreted as *definite* and *specific* in reference. Non-subjects are in principle dispensable. This applies to ergative and accusative in the same manner.

²³ Josef Perner’s principle of understanding: “*You haven’t understood anything properly if you haven’t understood why others haven’t understood it this way until now.*”

²⁴ There is a non-linguistic reason, on top of it: “*People tend to follow one another in what they say, do, and think. If there is a habit of doing things in a certain way, then people will do them in this way. So it is with language description.*” Dixon (2010: 66. vol.1), and he is right.

²⁵ “*This opinion was assisted by the fact that most ergative languages then known were spoken by small communities outside the mainstream of European culture (even the Basques were characterised in this way).*” Dixon (1984: 214).

Let us finally note that things fall in place properly when the absolutive (and definitively not the ergative) is identified as the counterpart to the nominative as the grammatical subject case. Additional empirical details that support this conclusion are systematically listed and discussed in the following section.²⁶

3. “Subject” as the grammatically privileged core argument

Grammatical-subject “privileges” manifest themselves in various grammatical dimensions, such as morpho-syntax or phrase structure, or even in prosodic phonology, as Dimmendaal (1983: 251) pointed out for the tonal marking of absolutive in Nilotic languages such as Turkana, or Yu (2021) for Samoan. In languages that provide a unique structural subject *position*, the subject is phrase-structurally privileged, and this is directly reflected in the structural hierarchies in a clause, that is, in c-command relations. In genuine [S[VO]] languages, that is, languages with a structurally defined preverbal subject position outside of the VP, this position is reserved for the subject while objects follow the verb within the VP.

The properties listed in Table 1 are typical manifestations of the privileged status of grammatical subjects in the syntactic literature. Even if it is not exhaustive, the list is sufficiently long. In view of the list in Table 1, it can hardly be denied that the grammatical patterns of the absolutive and the nominative settings are matching. The absolutive (but not the ergative) behaves like the nominative.

	Table 1: Grammatical privileges of subjects	Nom-Acc	Abs-Erg
a.	indispensability	Nom	Abs
b.	omission obligatorily grammatically signalled	Nom	Abs
c.	agreement with the finite verb in languages with agreement	Nom	Abs
d.	target of a ‘promoted’ NP with dependent case	Nom	Abs
e.	default structural case in case languages (finite clauses)	Nom	Abs
f.	not lexicalized in non-finite clausal constituents	Nom	Abs
g.	dropped if pronominal in null-subject languages	Nom	Abs
h.	top accessibility in languages with accessibility restrictions	Nom	Abs
i.	SVO ²⁷ order type: preverbal argument is the subject	[Nom [VX]]	[Abs [VX]]

Viewed from the vantage point of the old-school doctrine, Table 1 depicts a mysterious parallelism between the *subjects* of nom-acc languages and the *objects* of sentences in ergative languages. Those who like mysteries can stick with it. My preferred company is people who want to untangle them. And indeed, the mysteries disappear as soon as one admits that the grammatical characteristics of absolutives are typical of grammatical subjects. After all, this change in thinking almost certainly requires less mental effort than a Copernican shift from the Earth to the Sun as the centre of celestial movements, as the following discussion of the nine grammatical features in Table 1 will hopefully show.

²⁶ The following section is a modified version of Section 2.2 of Haider (2021b).

²⁷ In [S[VO]] languages, the subject precedes the VP, which contains the phrase-initial verb and all other core arguments. In SOV and VSO languages, any argument of the verb remains inside the domain of the verb, that is, the VP. The verb is phrase-final or phrase initial, respectively. An immediate consequence is the absence of the SVO-typical structural *subject-object asymmetries* in VSO and SOV (Haider 2013: 10-11;).

3.1 Indispensability. As known at least since Keenan (1976), nominative is mandatory while accusative is optional for many verbs, as in “unspecified-object”²⁸ constructions. The same is true for absolutive and ergative arguments, respectively. The grammatical source of this restriction is obvious. It is the dependency relation between subject status and dependently case-marked object-status: Omit a dependently-case argument and no grammatical relation changes (1a). Eliminate the subject in a nom-acc language, and the object becomes the subject (1b). The ergative can be eliminated freely (1c,d). However, as illustrated by Polinsky (2013): WALS, chapter 108,²⁹ if the *absolutive* argument is eliminated, the *ergative* becomes *absolutive*,

- | | |
|--|--|
| (1) a. The subject precedes (objects). | Active: <i>optional accusative</i> |
| b. Objects are preceded (by the subject). | Passive: obligatory direct object-to-subject |
| c. na’e tamate’i (’e ’tevitā) ’a koliate. | Tongan (Churchward 1953: 68). |
| has killed (ERG-David) ABS-Goliath | |
| ‘David killed Goliath’ | [without Erg: ‘Sb. killed Goliath’] |
| d. (Unai-k) pakete-a-k bidal-i ditu. | Basque (Fernández & Berro 2022: 1050) |
| (Unai- ERG) package-det-pl[abs] send-pfv have.3plAbs[3erg] | |
| ‘Unai has sent the packages.’ | [without Erg: ‘Sb. has sent the packages’] |

The indispensability is a direct reflex of the case-dependency relation between accusative and nominative on the one hand, and ergative and absolutive on the other hand. The optional omission of the superordinate case (nominative, absolutive) would completely obscure the subject-object identification because it would trigger the case-switch of the dependent-marked argument. Morpho-syntactically, nominative and absolutive behave in the very same way.

3.2. The antipassive myth

That is the biggest myth of all. Wouldn’t it be mysterious if there were languages that employ explicit grammatical means for eliminating a *direct object*, but their subjects may be dropped freely. Second, as mentioned in the introduction, it must trigger a warning reflex in every grammarian’s mind when (s)he hears that there are languages in which a subject in a finite sentence seems to be systematically degraded to an object when the object is omitted. That’s too bizarre to be taken at face value. If this were a legitimate grammatical option, it shouldn’t be limited to ergative languages, but it evidently is, and it is known as “antipassive”.

This myth goes hand in hand with another assumption, the functionalist belief in *grammaticalised* pragmatic effects such as foregrounding and backgrounding. Supposedly, the function of the passive voice is to background the default subject of a transitive verb, while the antipassive voice backgrounds³⁰ the default object argument.³¹ What is overlooked is the fact that this is merely the description of a collateral effect, but not the grammatical causality, although unyielding functionalists seem to foster the opposite view.

²⁸ He ate (*something*). He hit at *me* (vs. He hit *me*).

²⁹ <https://wals.info/chapter/108>

³⁰ “For these constructions with an antipassive effect, it is the patient that is backgrounded.” Mithun (2021: 56-57).

³¹ “Approaches have focused primarily on the patient when identifying antipassive constructions. Vigus considers an antipassive to be any construction where the patient is of lower topicality than in the corresponding basic voice construction. Polinsky notes that while there is no single morphological or discourse indicator of the antipassive cross-linguistically, the demotion of the semantic patient is a diagnostic feature.” Heaton (2020: 135).

Typologists who endorse a crude functionalist stance must face the question of how such a sensational anthropological schism could have come about. Apparently, people who speak one type of languages only stare intensely at their subjects and overlook their objects, while the opposite is true for people who speak the other type of language. Typical ergative languages function well without a passive voice, while typical nom-acc languages have no need for a systematically operating antipassive. No one can seriously believe that speakers of ergative languages are stubbornly fixated on what is affected by an action, while speakers of accusative languages are obsessed with who is in control of the action. That would be material for satire, rather than for science.

What is most disturbing about this is the fact that a majority of functional typologists do not seem to be concerned about the situation. The virtually complementary distribution of passive and antipassive as grammatical voice systems is another blind spot in probability judgements. What we see here is in fact a showcase of the unholy alliance between a grammatical bias and an unreflective functionalist creed.³² An unbiased view without ideologically motivated constraints, opens up a straightforward path:

Grammatical subjects, unlike objects, must not be omitted, unless signalled morpho-syntactically. In nom-acc languages, such grammatical signals are known as "passive voice" and as "antipassive voice" in ergative languages. In reality, passive and anti-passive are just two sides of the same coin. They are the grammatical signal for the omission of the default subject candidate, which covaries with the type of alignment. As a consequence of case dependency, the argument with dependent case surfaces as nominative (by acc-to-nom) or absolutive (by erg-to-abs), respectively. This complementarity is reflected in the cross-linguistic distribution.

Given these circumstances, it is striking that for Polinsky (2013), it is not worth mentioning that among the twenty-one languages³³ with "productive" antipassive voice that she lists, there is not a single nom-acc language. On the other hand, she does not mention that the passive voice, as a grammatically coded diathesis, has no systematic place in the grammar of ergative languages. The complementarity is predicted if passive and antipassive are different names for the same grammatical device, namely the blocking of the default subject argument of the verb.

Of course, the characterization of antipassive as subject-blocking device does not contradict the fact that, in addition, there may be diverse and partially idiosyncratic grammatical means to block the direct object (= argument with the dependent case). However, these are phenomena that should not be confused with the ergative *passive voice*, aka antipassive. Collateral damages of the functionalist stance come from poor eyesight in this respect. Apparently, anything that

³² Functionalism as an *explanatory* framework is invalid (Haider 2015a: 205-220). A functionalist 'explanation' is not a scientifically valid explanation. Functions are not the *causes* of forms. The appeal to (e.g. communicative) functions and functional profiling is a typical common sense reaction, as we humans prefer "final causes" (see Aristotle) as an answer to our favorite question "What is the use and purpose of this?" Functional answers to "How does it work?" would be legitimate, but not as answers to "Why is it there?" See Haider (2021a: 110-112) and the arguments of philosophers of science cited there. Functional design of grammars is a result of persistent variation & selection, that is, *cognitive evolution* of grammars as cognitive systems (Haider 2021a).

³³ Acoma, Chamorro, Chukchi, Comanche, Diyari, Dyirbal, Godoberi, Gooniyandi, West Greenlandic, Halkomelem, Hunzib, Jakaltek, Kiowa, Koyraboro Senni, Krongo, Lango, Mam, Ojibwa, Tzutujil, Warrungu, Zoque.

looks like a fish is considered a fish, and anything that eliminates the patient-denoting argument of a transitive verb is regarded as an example of ‘anti-passive’.³⁴

Such an attitude does not reflect the current state of knowledge and, with all due respect, is syntactically uninspired. It must not come as a surprise therefore that “*antipassive functions are more varied than often described, and do not necessarily correlate with morphological or syntactic ergativity.*” (Heaton 2020: 131). People who claim to have discovered an anti-passive construction in a nom-acc language are syntactically blind in at least one eye. What they overlook is that antipassive triggers a case switch. Ergative switches to absolutive and therefore, correspondingly, nominative should switch to accusative. This is pure fantasy, of course, but it shows that there is a lack of insight into structural symmetries.

The result of comparing things that don't belong together grammatically is often “more varied” and inconsistent.³⁵ Passive and antipassive are grammatical voices. Grammatical voice is understood here as a general grammatical category, “*formally explicitly marked*” whose particular value corresponds to particular operations on the syntactical realisation of argument structures. Antipassive, as characterised by Dixon (1994:146) and Dixon & Aikhenvald (2000:9) is an exemplary case of voice:

- (2)a. The *antipassive* construction is formally explicitly marked.
- b. *Antipassive* forms a derived intransitive³⁶ from a transitive verb.
- c. The otherwise *ergative*-marked NP becomes S (viz. subject).
- d. The otherwise *absolutive*-marked NP goes into a peripheral function and can be omitted.

The four properties characterize the grammar of subject elimination in an abs-erg system. Please note that modulo alignment, (2) translates one-by-one into nom-acc passive, see (3). In an abs-erg system, the so-called antipassive is what passive is in a nom-acc system.³⁷ In fact, antipassive and passive do not deserve to be terminologically separated. They are instantiations of the *very same* thing, namely the signal of the omission of the default subject argument in different alignment systems. “Antipassive” is the passive of ergative languages, nothing more and nothing less.

- (3)a. The *passive* construction is formally explicitly marked.
- b. *Passive* forms a derived unaccusative from a transitive verb.
- c. The otherwise *accusative*-marked NP becomes S (viz. subject).
- d. The otherwise *nominative*-marked NP goes into a peripheral function and can be omitted.

An immediate consequence is the familiar object-to-subject switch of the dependent argument, namely ergative-to-absolutive and accusative-to-nominative, respectively, in the antipassive

³⁴ Antipassive, understood as a conceptual umbrella for *any* phenomenon in *any* language in which the theme argument of a transitive verb is neutralized, is not a grammatical voice, but only a misleading functionalist concept. Biologists don't do this. They consider it misguided to sort organic systems according to function.

³⁵ Zoologists would not organize conferences on *the rear end of water dwelling vertebrates*, discussing the parametric difference between horizontal and vertical tail fins of pisces (fish) and catacea (whales, dolphins, etc.), respectively. Functionalist schools in linguistics do not hesitate to compare incomparable things, provided they only appear to have the same lifelike function (see: cross-linguistic collections of ‘antipassive’ phenomena).

³⁶ The appropriate term for the derived antipassive form of the verb is not ‘intransitive’ but *unergative*, since the ergative-marked argument of the active construction switches case and surfaces as absolutive.

³⁷ “*Ergative systems have an analogous construction, here termed as antipassive, which has all the properties of the passive, as Kuryłowicz, again saw.*” Silverstein (1976: 140), who is the namesake of ‘antipassive’.

and the passive of languages with a structural case system. This is the situation in grammars in which a subject is grammatically defined.³⁸

The current confusion with passive and anti-passive is largely due to a misguided perception, namely the confusion of cause and effect. The cause is the obligatory signalling of the omission of the primary subject candidate. The effect in nom-acc systems is the elimination of the agent argument while in abs-erg systems the eliminated subject candidate is the ‘undergoer’ argument.

Functionalists who focus more on the communicative effect than on the grammatical causalities have fallen into their own trap. Polinsky (2013) and (2017a: 310), describes passive and anti-passive in the functional way of focusing on a circumstantial property, namely *"In the passive, the suppressed or demoted argument is the agent-like argument, in the antipassive, the patient-like argument"* and antipassive (2017a: 310) is a clause *"with a transitive predicate whose logical object is demoted to a non-core argument or non-argument"*.

Let me emphasise once more that such a characterisation is both misleading and missing the essential generalization. The syntactically relevant property is *not* the specific semantic or pragmatic role of the ‘demoted’ argument or its logical argument content. Grammatically relevant is the *syntactic* role, and this role is the function of a *grammatical subject*. Whenever in a finite clause, the argument of the verb that would otherwise surface as a syntactic subject, viz. as *nominative* or *absolutive*, respectively, is syntactically eliminated, this must be formally marked morpho-syntactically. The naming of the semantic role of the suppressed subject is redundant since it is determined by the alignment system in interaction with the lexical argument structure. “Demotion of the logical³⁹ object” is a misleading concept. It cannot satisfactorily answer the question of why antipassive voice and passive voice are in complementary distribution. In fact, no object is demoted in antipassive; a primary subject candidate is eliminated, and the dependent case is replaced by the subject case, that is acc-to-nom or erg-to-abs, respectively.

The missed cross-linguistically valid generalization is this. Ergative and accusative are dependent cases and behave alike: When the candidate for the superordinate case – nominative or absolutive – is eliminated, the noun (phrase) with dependent structural case switches into the superordinate case. It is ‘promoted’. The promotees are accusative and ergative, respectively, but crucially not the absolutive, which ought to be promoted if it were the alleged object.

Relation changing is asymmetric. There is no language that ‘demotes’ a subject to the object function whenever the object has been eliminated. In other words, no language turns a nominative into an accusative when the candidate for accusative is absent, and by the same token, no abs-erg language would replace an absolutive by an ergative case when the candidate for the ergative case is absent. The dependency works in the opposite direction, and it is perfectly captured once absolutive is recognized as the case of the subject. Note that this is a *syntactic* property. The usage of the term “morphological ergativity”⁴⁰ seems to insinuate that this is a property of morphology. This is not so, of course (see section 3.10). It is a core issue of the syntactic

³⁸ It crucially does not hold in grammars that do not define a grammatical subject, such as tripartite or stative-active systems, in Comrie’s (2005) terminology; See Mithun (2005, 2008) for an informed discussion and data.

³⁹ There is nothing ‘logical’ about a ‘logical object’. In predicate logics, there are no subjects or objects, but only arguments. Aristotelian terminology confuses more than it clarifies.

⁴⁰ Dixon (1994:39;143) uses “morphological ergativity” for referring to intra-clausal relations and “syntactic ergativity” for inter-clausal relations (“syntactic pivots”).

alignment system that is reflected in grammatical morphology. Morphology codes syntactically determined information. Anyone who casually refers to ‘morphological ergativity’ should also consider the opposite perspective: What would be morphological vs. syntactic “nominativity”?

3.3 Subject-verb agreement

Agreement between subject and finite verb is a widespread morphosyntactic phenomenon. The verb agrees with the subject, and if a language requires object-agreement for the finite verb it also requires subject-agreement. The only exception would be ergative languages in the traditional view, which regards the absolutive as an object case and not as the general subject case. In this case, agreement with absolutive and not with ergative falsely appears to be object agreement in the absence of subject agreement.⁴¹ However, this is only an illusion from which one is immediately cured when one steps out of the myth and realises that ‘absolutive’ is the subject case.

3.4 Target of a ‘promoted’ NP-argument (object-to-subject)

Object arguments that are grammatically turned into subjects are a cross-linguistically frequent phenomenon. In nom-acc languages, the object argument appears in the subject case and in the subject position, if there is one. Proto-typical cases for promotion are passive, middle, or verbs with an argument structure that contains only an object argument but no subject candidate. In nom-acc languages, such verbs are known as *unaccusative* verbs. Their counterparts in abs-erg languages are ‘unergative’ verbs. This needs to be explained.

As illustrated in (4), the intransitive argument-structure format is the transitive minus the argument in the dependent case. This is nominative without accusative argument or absolutive without ergative argument. The unaccusative format, on the other hand, contains a would-be accusative that surfaces as Nominative (= subject). Correspondingly, the unergative format contains a would-be ergative that surfaces as absolutive (= subject). The underlined argument A is the argument to be assigned the superordinate case, viz. nominative in (4a) and absolutive in (4b).

- (4) a. Nom-Acc: transitive <A_i, A_j>, intransitive <A_i>, unaccusative <A_j>
 b. Abs-Erg: transitive <A_i, A_j>, intransitive <A_j>, unergative <A_i>

Unaccusative verbs are well-studied and described as verbs in nom-acc languages whose subject behaves in many respects like a passive subject, that is, a default object argument in subject function. Consequently, the absolutive subject of an unergative verb in an ergative language is predicted to syntactically behave in many respects like an antipassive subject, that is, a default ergative argument that surfaces as absolutive in subject function. This is an easy and clear prediction but it is hard to test without in-depth grammatical investigations of ergative languages.

3.5 Default and unmarked structural case (of finite clauses)

That nominative and absolutive are the unmarked cases in the respective case systems is an uncontroversial fact. “In an ‘ergative’ system the unmarked case, absolutive, almost always has zero realisation or at least a zero allomorph. Similarly, it is nominative that most frequently has zero realisation, or a zero allomorph, in an ‘accusative’ system.” And no languages are known

⁴¹ ChatGPT is a locus of *conventional* wisdom. It names Basque as an ‘exception’ to the generalization, referring to object agreement with the absolutive. However, Basque is an ergative language. Agreement with absolutive is subject agreement, of course, rather than object-agreement.

“*where ergative has zero form and absolutive is non-zero.*” Dixon (1994: 11). The obvious reason for this similarity is their shared grammatical function, namely that of a grammatical subject. Nominative and absolutive are default cases as well. A noun(phrase) in contexts that do not provide a case relationship, for instance in quotation form, appears in the unmarked, that is, nominative or absolutive form, respectively.

3.6 Case that is not lexicalized in non-finite clause

In ergative languages, *absolutive* case is not licensed in non-finite clauses, as for example in Dyirbal, Seediq, or Sama Southern (Deal 2015). For objects, such a restriction would be unheard of (cf. Aldridge 2008). As potential counterevidence, Polinsky (2017b: 17) notes that “*many ergative languages, including Inuit, have the absolutive freely available in non-finite clauses*” and that “*the so-called contemporative form of the verb in Inuit/Inuktitut is characterized by agreement with the absolutive but not the ergative.*”

The second part of the quote is the key to the first. The very same constellation is found in Nom-Acc languages, too. Portuguese is the Nom-Acc counterpart of Inuit, with nominative occurring in the so-called inflected-infinitive construction, cf. Madeira & Fiéis (2020). Another example is Hungarian (Tóth 2020). In each case, *agreement* licenses the assignment of subject case, in finite as well as in infinitival context, since both contexts provide an inflected verb with agreement features. The typical infinitival clause without a lexical subject is a clause *without* subject-verb agreement. Lack of agreement is the factor blocking a subject case.

3.7 Dropped as pronominal in null-subject languages

In null-subject languages, the case of the null-subject pronoun is nominative (in the nom-acc setting) or absolutive (in the abs-erg setting). Examples of the latter come, for instance, from Basque or Hindi. The literature on ergative pro-drop is deficient, though. The optional presence/absence of the ergative (with an interpretation of existential closure) tends to be mistaken as a case of pro-drop, which it is not. It is necessary to clearly differentiate between the genuine *referential* pro-drop of the absolutive (= definite specific reading) and the free omission of an ergative as a dependently cased argument (with indefinite existential reading).

3.8 Top accessibility in languages with accessibility restrictions

A reliable indicator of a syntactic subject function is Keenan & Comrie’s (1977, 1979) accessibility hierarchy. No languages are known in which the grammatical subject would be generally inaccessible for relative clause formation, but there are many languages in which *only* the subject is accessible. For Polinsky et al. (2012: 69). “*Ergative languages have posed challenges to the AH [accessibility hierarchy]_{HH} in that many of them exhibit syntactic ergativity: In many of them, the absolutive arguments (intransitive subject and transitive object) relativize with a gap, but the ergative DP does not.*” Evidently, the absolutive is not the case of the object. It is the case of the grammatical subject. Enigma dispelled.

3.9 SVO-order type: preverbal subject and postverbal object

Linguists of two schools that often look down on each other, that is, functional typologists and generativists, are united in their claim that “ergative SVO languages” do not exist. This is an immediate and wrong consequence of their empirically inadequate concept of grammatical subject. Typologists [Siewierska (1996)] and historical linguists [Trask (1979)] put it on the list of

properties of ergative languages while generativists – Mahajan (1997), Lahne (2008), Taraldsen (2017), Roberts (2021) – eagerly try to reconcile their axiomatic theory with this alleged fact. Viewed from a distance, it would be truly surprising if one of the three major types of sentence organisation were inaccessible for ergative languages. It is NOT. Again, it is merely a self-inflicted mystification.

For these authors, “*no ergative SVO*” means that there are no languages with an [Erg [V Abs]] clause structure. However, this is *not* the clause structure of an ergative [S[VO]] language. Ergative SVO clauses are [Abs [V Erg]]. On the other hand, it is almost trivially true that [Erg [V Abs]] languages must be inexistent; there are no [Acc [V Nom]] languages either. A non-subject in the typical structural subject position of an [S[VO]] clause structure and the subject in a VP-internal position is no admissible *canonical* base structure of clauses in any language.⁴²

Those who think the absence of [Erg [V Abs]] needs attention identify ergative case with subject case. Ergative SVO languages *do* exist and they are attested, but incognito. Languages that have uncontroversially been classified as (ergative) OVS languages, in fact, have to be re-classified as “SVO” languages (see sect. 4). These are [Abs [V ERG]] languages. The preverbal, non-agentive noun phrase is not the syntactic object. It is the syntactic subject of an abs-erg-language. The subtle point is harmless for the linguistic description of an individual language but the subsequent typological interpretation, that is, the step from “agent” or “patient” to “subject” and “object”, respectively, is detrimental if “ergative” is equated with “subject case”.

In other words, there is no justification for classifying an *ergative* language as “OVS” whenever its obligatory serialization pattern in simple clauses with non-pronominal noun phrases happens to be [Patient [V Agent]]. This, however, is exactly what happens in typological surveys, as for instance in WALS (Dryer & Haspelmath 2013) and others. Pāri, for instance, is listed as “OVS” (feature 100A) and “ergative” (feature 81A) in WALS, although in an earlier publication, Dryer (2007: 70) himself has explicitly qualified such a classification as “*somewhat misleading*.”⁴³

3.10 Subject properties of ergatives?

A treatise on the subject properties of the absolutive would be incomplete without going into the alleged subject characteristics of ergative noun phrases. After all, it is the defensive bastion of the advocates of the traditional view. Anderson (1976) has pointed out, and Dixon (2010a: 229)⁴⁴ repeats it, namely, that the ergative-marked argument is found also in contexts, which are the domain of nominative subjects in nom-acc languages, such as with imperatives, the binding of reflexives, or as subject controllers in infinitival constructions. A metaphorical reply is in place here again. Not all that glitters is gold. The characteristics mentioned are typical of subjects but are not exclusive characteristics of subjects. Here are the reasons.

⁴² The research literature is necessarily silent on the issue because of misclassified “OVS” languages. As will turn out, the absence of [O[VS]] is a major typological issue, too. A plausible explanation seems to be the following. The structural positioning is *congruent* with grammatical dependency relations. The dependent item (see dependent object case) is c-commanded by the superordinate item; hence the subject c-commands the object in the base configuration which it would not in an [O[VS]] base structure.

⁴³ In Dryer (2007: 70): “*Characterizing such languages as OVS is somewhat misleading in that the word order really follows an ergative pattern Abs-V-(Erg).*”

⁴⁴ “*Even in ergative languages, S and A share a number of properties – as addressee in imperative constructions, as controller of reflexive, and so on*”.

Let us start with **imperatives**. What would an imperative mean that obligatorily addresses the *non-agentive* argument of an agentive transitive verb, that is, an absolutive subject? When using an imperative, we directly address a potential agent. If the grammar of an ergative language would indeed require to relate the imperative to the absolutive, the message would not be delivered to the correct addressee. One can be sure that the evolution of grammars will have led to systems that provide the appropriate option irrespective of the alignment system. Imperatives with the absolutive argument of a transitive verb as referent would be dysfunctional. In nom-acc languages as well as in abs-erg languages, the addressee is the relevant argument role for imperatives, irrespective of the case marking rules. Note the suddenly structural argumentation of functionalists here.

Reflexives in the function of core arguments represent a similar configuration. Logically, in a reflexive predicate, the variables of at least two argument positions are identical. The two variables of (5a) are satisfied by the same constant in (5b), as exemplified in (5c).

- (5) a. [HURT (x,y)] & reflexive: $y = x$. Therefore: [HURT (x,x)]
 b. Donald hurt himself (this week).
 c. [HURT (d,d)], [[Donald]] = d

Let's now reflect for a moment on the relationship between the logical form and its linguistic rendering in reflexive constructions. Cognitively, the semantics of transitive verbs establishes a causal relation: An agent does something and the one affected by this action is identical with the agent. If in an ergative grammar, the participant that binds the reflexive is the ergative, then the causal *agent* of the denoted situation is introduced immediately. Now imagine that the absolutive were the *obligatory* binder of a reflexive in such a language. In this case, we would be told by an equivalent of (5b) that something happened to Donald_{Abs}, and that someone hurt him, and only then the reflexive indirectly supplies the information on the agent, namely “by himself”.⁴⁵ The direct way with ergative as the binder seems to be the straightforward one. A parallel situation exists in nom-acc languages. The direct object can bind a reflexive pronoun if the semantics require it, as in (6a), or (6b) with reciprocals. English is fully representative in this respect.

- (6) a. When squaring, you have to multiply *a number* by *itself*.
 b. He introduced *them* to each other – *They* introduced him to each other.

What we see when we take off ideological blinders is that the grammars of human languages do not generally serve to confuse the listener. In semantics, reflexives serve as bound variables and variable binding is not obstructed by grammar as long as it respects the structural prerequisites, viz. c-command.

The mental computation of linguistic in- and output takes into account both the syntactic structure and the lexical-semantic argument structure. Manning (1996a, 1996b) argues that binding data are no reliable indicator of syntactic subjecthood of ergative noun phrases. The construal process operates on the one hand on the information provided by the lexical argument structure (w.r.t. the selection of the binder) and on the other hand on the syntactic structure (w.r.t. to c-command of the binder). The ergative may bind if it c-commands. Analogously, the accusative

⁴⁵ This is not excluded, as “Donald could have been hurt by himself” shows. A Google search produces about two million hits for “hurt himself”, but only a quarter of a million for “hurt by himself”.

is able to bind if it c-commands (7a). In (7b), only the subject would be able to bind.

- (7) a. Dass ich beim Quadrieren *jede Zahl* [mit sich] multiplizieren muss (German)
 that I at squaring each number_{Acc} [with itself] square must (= 6a)
 b. *Dass ich beim Quadrieren [mit sich] *jede Zahl* multiplizieren muss
 that I at squaring [with itself] each number_{Acc} square must

If an absolutive does not anaphorically bind an ergative reflexive, as Polinsky et al. (2012: 268) emphasizes, this is not only an asymmetry in terms of phrase structure but crucially also one in terms of argument structure. The latter asymmetry is as relevant as the former. In sum, neither imperatives nor reflexives provide compelling evidence for the subjecthood of ergatives in ergative languages.

In **infinitival constructions**, the situation is similar, but it is complicated by the fact that there are two kinds of infinitival construction, namely clausal complements as well as mono-clausal clause-union constructions. This is easier to illustrate by German than by English because in German both constructions coexist for a sizeable class of verbs, see Haider (2010, Ch.7) and (2013:86-87). (8a) is the clausal-complement construction, with the silent, controlled pronominal subject. (8b), however is the mono-clausal verb-cluster construction. It is absent in SVO languages, for principled reasons (see Haider 2010). In the cluster, the argument structures are combined to the argument structure of the cluster.

- (8) a. dass sie [PRO es *nicht* anzufassen]_{CP} [hätte{beginnen/... } können müssen]_{verb-cluster}
 that she [it not to-touch would-have {begin/....} can must
 b. dass sie es *nicht* [hätte anzufassen{beginnen/versuchen/vergessen} können]_{verb-cluster}
 that she it not [would have to-touch{begin/ try/ forget} can
 c. [Anzufassen {beginnen/... } müssen] hätte sie es nicht
 to-touch {begin/ ...} have-to would-have she it not
 d. dass ihm [zu grauen begann]
 that him to dread began
 e. *Es ist möglich, [PRO ihm zu grauen] – Es ist möglich, dass ihm graut
 It is possible, [him to dread] – It is possible, that him dreads

The clausal infinitival has an obligatory silent subject (8a). If there is none, the infinitival clause is ungrammatical (8e). The cluster-construction may be subjectless, depending on the base verb (8d). The verb *grauen* (dread) has a single argument, which is a dative, and the verb *beginnen* (begin) is optionally subjectless. The *clausal* construction is ruled out for *grauen*, since there is no argument for the subject function of the infinitival (8e), the clause-union construction (8d) is grammatical.

What is the implication for ergative languages: Many of them are OV language with V-clustering. So, the argument-pooling can include the ergative argument as well. At the level of the lexical argument structure, that is, in the clause union construction, the agent argument may function as pivot with respect to control, binding, and imperative constructions, as Manning (1996a,b) suggested. Finally, it should be noted that a control relationship in an infinitive construction is not conclusive evidence for the existence of a subject relation. A nom-acc language such as Icelandic shows that a dative argument binds a reflexive or represent the subject of an infinitival construction if it is in the structural subject position, see Thráinsson (2007: 420).

4. "OVS" languages are SVO languages with ergative alignment

This section⁴⁶ shall succinctly demonstrate that virtually all of the hitherto undisputed candidates for the category "OVS language" are Abs-V-Erg languages. The syntactic subject of an ergative language is the *non-agentive* argument, that is, the so-called patient-argument. Hence, in (syntactic) reality, an ergative "OVS" language is an SVO language with Abs-Erg-alignment. The source of the present misperception is the non-structural characterization of grammatical functions, namely the equivocation of a lexico-semantic stereotype, viz. agenthood, with "syntactic subject".

It will turn out that in most typological surveys, P-V-A is counted as "OVS" without taking into consideration the particular alignment system of the given language, which is almost always ergative. If type-assigned correctly, these languages have to be registered as SVO languages with ergative alignment. Dixon (1994: 50) explicitly notes that for "*languages with syntactic function shown by constituent order*" SV/OVA is a sign of ergativity.

Greenberg (1963: 76) describes OVS as one of the types that "*do not occur at all or, at least are excessively rare*", and this has proven correct, contrary to positions held in the typological literature. Greenberg's (1963) original sample of thirty languages contained only two languages classified as OVS, with VOS as alternative word order, namely Siuslaw and Coos (s. Greenberg's Appendix II). Both languages are ergative; see Mithun (2005), Frachtenberg (1913: 128, 154). In a survey, Hammarström (2016: 25) calculated the constituent orders of 5252 languages partitioned into 366 language families. His count yields 40 OVS languages (out of 5252), belonging to three languages families. The corresponding percentage is 0,07% of the total number of languages.

Dixon (1994: 50-52) itemizes the following ergative languages as instances of SV/OVA, that is, *ergative* "OVS" languages: Kuikúro, Macushi,⁴⁷ Maxakalí, Pări, and Nadëb. Further confirmation can be found on Kuikúro in Franchetto (1990, 2010), on Macushi in Abbott and Foster (2007) and in Carson (1982), on Maxakalí in Popovich (1986), on Pări in Andersen (1988), and on Nadëb in Martins & Martins (1999).

Dixon also refers to a second pattern, namely VS/AVO, and illustrates it with Huastec and Paumarí. Huastec is described as an SVO language by Edmonson (1988). It is a Mayan language which Edmonson (1988: 116, 570) describes as an ergative language, with the basic order A-V-O-IO. Her crucial sample, however, consists of exactly *five* sentences of a form in which *both* arguments of a transitive verb are present as full noun phrases. "*Sixteen clauses have a variant order (O TV, TV A, etc.)*" (Edmonson 1988: 568). Since Mayan languages are predominantly V-initial (England 1991), the Huastec data does not provide convincing evidence for a *basic* OVS structure.

Paumarí has been characterized as split-ergative language by Chapman & Derbyshire (1991: 267, 271) with ergative-absolutive for full noun phrases and nom-acc alignment for pronominal arguments and only the immediately preverbal noun phrase is case-marked. Chapman &

⁴⁶ This section is largely identical with Section 3 of Haider (2021b).

⁴⁷ Dixon (1994:138) classifies Macushi as ergative. Dixon (2010, vol.1: 73) criticizes Ethnologue: "Macushi [...] is given as OVS, despite the excellent grammar of this language specifying that the 'basic orders' are OVA (although AOV also occurs frequently) and SV."

Derbyshire (1991:164, 250) assign "SVO" as basic word order to Paumari. This deserves a comment, since in an *ergative* setting, "AVO" would *structurally* be OVS. On the other hand, the language has a passive construction, but no antipassive. Zwart & Lindenbergh (2021: 30) argue that when case is coded (viz. only in the preverbal position, S, A, and O are coded differently, which is a tripartite system. It does not qualify as an ergative language and consequently not as an [O[VS]] language.

In a study on word-order type and alignment, Siewierska (1996) lists four languages as "OVS" out of a sample of 237 languages, namely Macushi and Pări, as in Dixon's sample, plus Hixkaryána, and Southern Barasano. For the latter, Jones & Jones (1991) presented a syntax monograph that has been reviewed by Dryer (1994). He criticises their type assignment⁴⁸ and concludes: "*It is possible that it is best treated as indeterminately SOV/OVS, a word order type that appears to be quite common in the Amazon basin.* (Dryer 1994: 63). Hixkaryána will be discussed together with the following set of languages.

In WALS⁴⁹ (Dryer & Haspelmath 2013), the following languages are listed as "OVS". Four of them are plainly ergative, namely Kuikúro, Macushi, Pări, and Tuvaluan.⁵⁰ Four are caseless (i.e. 'neutral' alignment) but with ergative properties: Asurini, Selknam,⁵¹ Tiriyo,⁵² Ungarinjin.⁵³ According to Primus (1995:1089), "*The Tupi-Guarani languages Asurini and Oiampi have ergative marking in dependent clauses.*"

Four potential candidates of OVS languages have to be discussed in detail further, namely Kxoe, Cubeo, Urarina,⁵⁴ and Hixkaryána. For Kxoe, Fehn's (2015: 214) grammar of Ts'ixa (Kalahari Kxoe) is very explicit: "*There are three patterns available for transitive clauses: AOV, AVO and OAV, with the latter occurring less frequently than the other two. Although the dominant word order of the Khoe languages is thought to be AOV (cf. Heine 1976, Güldemann 2014), AVO is just as frequent.*" The type-assignment in WALS exclusively follows Köhler (1981).⁵⁵ In sum, Kxoe does not seem to qualify as a reliable testimony of OVS. So, we are left

⁴⁸ "A count of all examples in the grammar shows both SV and VS order common, with SV slightly more common, though numbers of examples cited in a grammar is a poor source of data. [...] If we interpret the notion of an OVS language as referring to clauses with a noun object and a noun subject (the standard usage in word order typology), it is not clear that Barasano qualifies." (Dryer 1994: 63).

⁴⁹ In the introduction to chapter 82 of WALS, Dryer (2013) writes: "There are also languages [...] in which the order can be described as Absolute-Verb-Ergative: these languages are shown as OVS on Map 81A and as SV on this map. In fact, three of the six OVS languages shown on Map 81A are of this type: Pări (Nilotic; Sudan; Andersen 1988), Mangarrayi (Mangarrayi; northern Australia; Merlan 1982) and Ungarinjin (Wororan; north-western Australia; Rumsey 1982)."

⁵⁰ Besnier 1986: 245: "Despite the word-order freedom exhibited by Tuvalan, there is a basic order, and this order is verb initial." Besnier (2000: xxiv): "Case marking follows an ergative-absolutive pattern".

⁵¹ "Selk'nam seems to be an ergative language as to word order and verbal marking. Nevertheless, case marking is still an issue that remains to be debated, since the data now available is not sufficient to determine the typological nature of the language, which appears to have been an S marking/A-O unmarked language till the beginning of the twentieth century." Rojas-Berscia (2014: 23).

⁵² Rill (2017: 430): "In the end, Tiriyo verb agreement is best analyzed as ergative in alignment."

⁵³ Rumsey (1982:145) summarizes the "ordering norms": S precedes V, O precedes V, while A follows. This is exactly the order one expects to find if a language is an SVO language with ergative alignment.

⁵⁴ A text count based on 445 main clauses sampled from seven texts produced the following frequencies:⁵⁴ 3% OVA and 4% AOV orders (Olawsky 2006: 653; 2007: 45). 93% are clauses with null-subjects and/or null-objects. For dependent clauses, Olawsky (2006: 658) reports 0,3% VA and 0,8% AV orders.

⁵⁵ Köhler's reliability has been questioned: "He himself reduced the richness of Khwe cultural and linguistic expressions in his documentation by increasingly limiting field methods." (Boden 2018: 142).

with Cubeo, Urarina, and Hixkaryána.

As for Cubeo, according to the typological platform SAILS [South American Indigenous Language Structures, Muysken et al. (2016)], “*the patterns in the order of frequency in the data [are]_{HH}: OVS, SVO, VSO, SOV, and (least common) VOS* (M&M 1999:142).” WALS, which refers to the same source, namely Morse & Maxwell (1999), lists Cubeo as a nom-acc OVS language⁵⁶ that is generally head-final (postpositions, Gen-N, V-neg) and of the nominative-accusative type, with passive. This is crucial information for the analysis discussed below, since these properties are also shared by Hixkaryána and Urarina.

The essential issue to be settled for the three languages is this: Are these languages head-initial or head-final? If their VP is head-final, [OV] is likely to be a constituent. If they are head-initial, [VA] is a constituent preceded by O. The latter case would make them [O[VA]] languages, with “O” being the structurally highest argument in the clause. What are the relevant facts?

Cubeo, Urarina, Hixkaryána, are post-positional as well as Gen-N. According to Dryer (2007: 69) “*the fact that the characteristics in other languages pattern with the order of object and verb would lead us to expect both OVS and OSV languages to pattern with SOV languages. In so far as we have evidence, this prediction seems to be true. For example, Hixkaryana is post-positional and GN.*” The same is true for Urarina. In addition, as Kalin (2014: 1096) emphasizes, the adjective phrase is head-final, too. Olawsky (2006: 667-668) provides information on the V+Aux order of Urarina, an order that is completely absent in V-initial languages. Finally, Olawsky (2006: 662) notes that in negated sentences, AOV is an unmarked order, that is, A is not focussed. “*In a transitive clause, constituent order can be AOV as the result of negation.*” Taken together, these grammatical features are good indicators for a head-final organization of the verb phrase in the three languages.

The cumulative evidence for a head-final VP has lead Kalin (2014) to the conclusion, that Hixkaryana is an [[OV]S] language, with the VP⁵⁷ in a secondary, that is, fronted position.⁵⁸ This would support Derbyshire's (1981) conjecture that the OVS clause structure is the result of the loss of ergative case marking in the Carib languages. An [[OV]...S...] structure is the likely outcome when in an Abs-V-Erg system, case distinctions are lost and the alignment system is reinterpreted as nom-acc, while the word order is preserved. The result is a nom-acc system, with OVS order, at the price of a complication in clause structure by VP fronting. However, if the analysis of Kalin (2014) turns out to be robust enough, then we see a rare constellation of clause structure with a preposed head-final predicate phrase in these three languages.

Strong indirect support for such an analysis comes from Queixalós (2010:241, 254). He argues that the clause structure of Katukina-Kanamari is basically [[OV] S], and shows in great detail that this structure remains constant under *alternative alignments*. The frequent clause type is [[Erg V] Abs] and a less frequent one is [[Acc V] Nom]. Note that the arguments of the verb

⁵⁶ Ethnologue classifies this language as SOV. Wals follows the misleading maxim of classifying a language according to the most frequent pattern.

⁵⁷ “*Transitive clauses have a tightly bound OV verb phrase constituent that is usually followed by the subject NP. Des had actually said so in a dense 1961 paper I had not seen (IJAL 27, 125-142), packed with obscure formulae.*” (Geoffrey Pullum, Obituary: Desmond Derbyshire, *Linguist List* 19.1, Jan 03 2008).

⁵⁸ The same analysis, modulo head-initial V, is the standard structure of Malagasy, a head-initial language with VOS order. Its clause structure is taken to be [[V O] S], that is, with a clause-initial VP. (Paul 1999, Pearson 2001). Keenan & Manorohanta (2004: 178) formulate it as follows: S = [[Pred V^o(Θ) + (X)] + DP].

change places in the two clause types, but the structure of the clauses remains constant in terms of the formal grammatical functions, namely [[OV] S]. In the traditional terminology, one clause type would wrongly be classified as ergative “SVO”, and the other also wrongly as accusative “OVS”.

All in all, no language is known whose clause structure would be an instance of [O [VS]], which would be the structure of a genuine *structural* OVS language. There is no compelling evidence for a basic [O[VS]] clause structure.

As for ergative languages, the traditional “OVS” classification means Absolutive-V-Ergative order, and this is *subject-verb-object* order under ergative alignment. It is a consequence of the above discussion that the *structural* identification of grammatical relations is an indispensable basis for a cross-linguistically adequate comparison of clause structures.

5. Closing remarks

The theory just presented is a grammar-centred theory of ergativity. The old-school theory, on the other hand, is communication-centred, which is typical of functionalist approaches. The fundamental difference becomes clear in the answer to the following question. Do communicative functions determine the range of forms⁵⁹ (= functionalist), or does the range of forms determine what is available for communicative functions (= structuralist)? Grammatical forms are not *caused* by communicative functions; rather, grammatical forms are *used* for communicative functions. And what is it that is and has been coordinating these two domains? The cognitive evolution of grammars; see Haider (2015a, 2021a). The functional design of grammars is the result of hundreds of thousands of years of cognitive evolution of grammars. Random variation and constant but blind selection are two mechanisms of evolution which has led and still leads to a variety of adaptive designs in the absence of a designer.

What evolutionary quality lies behind ergativity? Why do we find more than one type of alignment across languages? The cognitive evolution of grammars is the answer. If the system potential has room for more than one outcome, there will be more than one outcome in the long run. Evolution is a dynamic process that opens up possibilities, not one that suppresses them. Which ones will spread is an orthogonal question, however.

The preceding step in the emergence of ergativity (and accusativity) is the emergence of order relations in case systems, with structurally conditioned cases and a dependency relation between them. This is the bifurcation point, namely the choice of the candidate for default case. This automatically means that the other structurally case-marked argument receives the dependent case. What determines the choice of default and thereby the superordinate case? Chance decides! If the higher ranked argument in the lexical argument structure ‘wins’, an accusative system is born. If, however, the more frequently used argument⁶⁰ ‘wins’, we are at the other end of the argument scale of the lexical argument structure, and the result is an abs-erg system. Both options are grammatically feasible, but the ergative option leads to a system with opposing

⁵⁹ „Range of forms“ is the abbreviation for morphosyntactic inventory, syntactic structures, and relations.

⁶⁰ A simple observation supports this assumption: Listen to real-life conversations of people cooperating in work situations: “What shall we do? Go to the river, choose a straight branch, sharpen it, find a good spot, and catch fish?” A verb, together with an argument (which functions as the subject in ergative languages and as the object in nom-acc languages), is a linguistically versatile unit in communication, even in a proto-language.

hierarchies. The selection of the subject begins at the lowest point of the thematic hierarchy, but the syntactic projection of the argument structure onto syntactic structures produces an opposite structure. This inherent tension is a reason for the unequal distribution of ergative and accusative languages, and the splits in ergative case systems.

Evolution is not a strategic planner, but rather a tinkerer, as Jaques Monod (“bricolage”) puts it. This means that ergativity can also come into being by accident in a nom-acc language (and vice versa). An example of this is the acquired ergativity of Indo-Aryan languages. They are nom-acc, except in the perfect tense, where they code ergatively. How did this happen?

Here is what happened. In most IE languages, the tense system has gradually changed to a so-called analytically coding system, which typically consists of an uninflected verb form plus an inflected auxiliary verb. In Romance and Germanic languages, for instance, the perfect is coded with a participle plus an auxiliary. For all verbs, except unaccusative verb, the auxiliary is typically either a former possession verb (*habere*, *avere*, *avoir*, ...) or a get-into-possession verb as e.g. the Germanic *have*, a cognate of Latin *capere* (grasp). If, on the other hand, the participle is combined with a *copula*, there is an unavoidable passive effect since the past participle form generally blocks the default subject argument. The ‘*have*’-type auxiliaries deblock. The copula-type auxiliary, unlike auxiliaries of the ‘*have*’-kind, does not deblock a blocked subject argument;⁶¹ see Haider & Rindler-Schjerve (1987). And that is the point at which the Indo-Aryan languages branched off. In these languages, no deblocking auxiliary has been (able to be) recruited. So, they had to make do with the copula.

As a consequence, the perfect tense worked like a passivising construction, with the default subject in an oblique case, although it was used as part of the tense paradigms of otherwise active constructions. Within a few centuries, this got streamlined into an ergatively-coding tense paradigm, with the direct object of the present tense clause as grammatical subject in the perfect tense, viz. absolutive. Please note once again that there can be no doubt that in ergative Indo-Aryan constructions, the absolutive is the subject case. This results from its origin in an originally passivising construction. The Indo-Aryan ergative case derives from the oblique case that is typical for Indo-European passive constructions, i.e. ablative or instrumental.

Summary: The grammatical elegance of ergative languages only becomes apparent when one does not read them against the grain. Once one has understood that the absolutive is the subject case, facts neatly fall into place.⁶²

Bibliography

- Abbott, Miriam & Patrick Foster. 2007. Macushi. In Mary R. Key (ed.), *Intercontinental dictionary series*. Leipzig: Max Planck Institute for Evolutionary Anthropology.
- Abbott, Miriam. 1991. Macushi. In Desmond C. Derbyshire and Geoffrey K. Pullum (eds.), *Handbook of Amazonian Languages* 3, 23-160. Berlin: Mouton de Gruyter.

⁶¹ Here is an example set from German (see Haider & Schjerve 1987):

- i. Der Junge *hat* ihm den Brief *zugestellt* – der *zugestellte* Brief – *der *zugestellte* Bote – der *zustellende* Bote
the boy has delivered him the letter – the delivered letter – *the delivered boy – the delivering boy
- ii. Der Brief *wurde* ihm (von einem Jungen) *zugestellt* (passive) – Der Brief *ist* *zugestellt* (adject. Passive)
the letter was him (by a boy) delivered the letter is delivered

⁶² The truly astonishing thing: In a “chat,” Google’s AI mode finally confirmed the above arguments. AI tools will soon qualify as journal reviewers, it seems.

- Aldridge, Edith. 2008. Generative approaches to ergativity. *Language and Linguistics Compass: Syntax and Morphology* 2(5): 966-995
- Andersen, Torben. 1988. Ergativity in Parí, a Nilotic OVS language. *Lingua* 75. 289-324.
- Anderson, Stephen R. 1976. On the notion of subject in ergative languages. In Charles Li (ed.), *Subject and topic*. 1-23. New York: Academic Press.
- Baker, Mark C. 1988. *Incorporation: A theory of grammatical function changing*. Chicago: The University of Chicago Press.
- Baker, Mark C. 1997. Thematic roles and syntactic structure. In Liliane Haegeman (ed.), *Elements of Grammar*. 73-137. Dordrecht: Kluwer Academic Publishers.
- Baker, Mark C. 2015. *Case: Its principles and parameters*. Cambridge: Cambridge University Press.
- Bakker, Dik & Anna Siewierska. 2007. Another take on the notion Subject. In Mike Hannay & Gerard J. Steen (eds.), *Structural-functional studies in English grammar*, 141-158. Amsterdam: Benjamins.
- Besnier Niko. 1986. Word order in Tuvaluan. In Paul Geraghty, Lois Carrington, & Stephen A. Wurm (eds.), *FOCAL I: Papers from the Fourth Int. Conference on Austronesian Linguistics*, 245-268. Canberra: Research School of Pacific and Asian Studies, Australian National University.
- Besnier, Niko. 2000. *Tuvaluan: A Polynesian Language of the Central Pacific*. London: Routledge.
- Bhatt, Rajesh. 2007. Ergativity in Indo-Aryan Languages. Retrieved (Oct. 15, 2025) from <http://people.umass.edu/bhatt/papers/mit-nov2007-handout.pdf>
- Boden, Gertrud. 2018. The Khwe collection in the academic legacy of Oswin Köhler: Formation and potential future. *Senri Ethnological Studies* 99. 129-146.
- Brewer, William & Bruce Lambert. 2001. The theory-ladenness of observation and the theory-ladenness of the rest of the scientific process. *Philosophy of Science*, 68. 176-186. doi:10.1086/392907.
- Carson, Neusa Martins. 1982. *Phonology and morphosyntax of Makuxi (Carib)*. Lawrence: University of Kansas PhD-dissertation.
- Casadevall, Arturo & Ferric C Fang. 2018. Elegant science. *mBio* vol. 9,1 e00043-18. doi: 10.1128/mBio.00043-18
- Chamoreau, Claudine. 2021. Morphological split ergative alignment and syntactic nominativeaccusative alignment in Pesh. *International Journal of American Linguistics*. 87(4): 501-537.
- Chapman, Shirley & Desmond C. Derbyshire. 1991. Paumarí. In Desmond C. Derbyshire & Geoffrey K. Pullum (eds.) *Handbook of Amazonian languages*, vol. 3, 161-352. Berlin: De Gruyter.
- Chomsky, Noam A. 1981. *Lectures on government and binding*. Dordrecht: Foris.
- Chung, Sandra. 2006. Properties of VOS Languages. In Martin Everaert & Henk van Riemsdijk (eds.), *The Blackwell Companion to Syntax*, 685-720. Oxford: Blackwell Publishing.
- Churchward, C. Maxwell. 1953. *Tongan Grammar*. London: Oxford Univ. Press.
- Comrie, Bernard. 1989². *Language universals and linguistic typology*. Chicago: University of Chicago Press.
- Comrie, Bernard. 2005. Alignment of case marking of full noun phrases. In Martin Haspelmath, Matthew Dryer, David Gil & Bernard Comrie (eds.), *The world atlas of language structures*, 398-403. Oxford: Oxford University Press
- Coon, Jessica. 2017. Two types of ergative agreement: Implications for case. In Claire Halpert, Hadas Kotek & Coppe van Urk (eds.) *Papers for David Pesetsky*. 361-370. Cambridge: MIT Working Papers in Linguistics.
- Craig, Colette G. 1977. *The structure of Jacalteco*. Austin: University of Texas Press.
- Croft, William. 2013. Radical construction grammar. In Thomas Hoffmann & Graeme Trousdale (eds.), *The Oxford handbook of construction grammar*, 211-232. Oxford: Oxford University Press.
- Davis Henry, Carrie Gillon & Lisa Matthewson. 2014. How to investigate linguistic diversity: Lessons from the Pacific Northwest. *Language* 90, e180-e226.
- Deal, Amy R. 2015. Ergativity. In Artemis Alexiadou & Tibor Kiss (eds.), *Syntax – Theory and Analysis*. (vol. 1), 654-707. Berlin: De Gruyter.

- Deal, Amy R. 2016. Syntactic Ergativity. Analysis and Identification. *Annual Review of Linguistics* 2. 165-185.
- Derbyshire, Desmond C. & Geoffrey K. Pullum. 1979. Object initial languages. *Working Papers of the Summer Institute of Linguistics*, Vol. 23, Article 2. DOI: 10.31356/silwp.vol23.02
- Derbyshire, Desmond C. 1979. *Hixkaryana*. Amsterdam: North-Holland.
- Derbyshire, Desmond C. 1981. A diachronic explanation for the origin of OVS in some Carib languages. *Linguistics* 17. 209-220.
- Dimmendaal, Gerrit J. 1983. Topics in a Grammar of Turkana. In M. Lionel Bender (ed.) *Nilo-Saharan Language Studies*. Michigan: African Studies Center: 239-271.
- Dixon, Robert M.W. 1994. Ergativity. Cambridge: Cambridge University Press.
- Dixon, Robert M. W. 2010. *Basic Linguistic Theory*, vol. 1-3. Oxford: Oxford University Press.
- Dixon, Robert M.W. 2020. *Basic linguistic theory*. vol. 1-3. Oxford: Oxford University Press.
- Dixon, Robert M. W. & Alexandra Y. Aikhenvald. 2000. Introduction. In Robert M.W. Dixon & A.Y. Aikhenvald (eds.), *Changing valency: Case studies in transitivity*, 1-29. Cambridge University Press.
- Dryer, Matthew S. & Martin Haspelmath (eds.). 2013. *The world atlas of language structures online*. Leipzig: Max Planck Institute for Evolutionary Anthropology.
- Dryer, Matthew S. 1994. Review of Wendell Jones and Paul Jones, Barasano Syntax. *Word* 45. 62-65. DOI: 10.1080/00437956.1994.12098343
- Dryer, Matthew S. 2006. Descriptive theories, explanatory theories, and Basic Linguistic Theory. In Felix Ameka, Alan Dench & Nicholas Evans (eds.), *Catching language: Issues in grammar writing*, 207-234. Berlin: Mouton de Gruyter.
- Dryer, Matthew S. 2007. Word order. In Timothy Shopen (ed.), *Clause structure, language typology and syntactic description*, Vol. 1, 2nd edition, 61-131. Cambridge University Press.
- Dryer, Matthew S. 2013. Order of Subject and Verb. In Matthew S. Dryer & Martin Haspelmath (eds.), *The world atlas of language structures online*. Leipzig: Max Planck Institute for Evolutionary Anthropology. (Available online at <http://wals.info/chapter/82>)
- Edmonson, Barbara Wedemeyer. 1988. *A Descriptive Grammar of Huastec* (Potosino Dialect). Ph.D. dissertation. New Orleans: Tulane University.
- England, Nora C. 1991. Changes in basic word order in Mayan languages. *International Journal of American Linguistics* 57. 446-486.
- Fehn, Anne-Maria. 2015. *A Grammar of Ts'ixa* (Kalahari Khoe). Unpubl. doct. diss. Univ. Köln, Inst. f. Afrikanistik.
- Fernández, Beatriz & Ane Berro. 2022. Basque impersonals in comparison. *Linguistics* 60: 1039-1101.
- Fleck, David William. 2003. *A Grammar of Matses*. PhD dissertation. Rice University.
- Frachtenberg, Leo J. 1913. *Coos Texts*. New York: Columbia University Press.
- Franchetto Bruna. 1990. Ergativity and nominativity in Kuikúro and other Carib languages. In Doris L. Payne (ed.), *Amazonian linguistics: Studies in lowland South American languages*, 407-428. Austin: University of Texas Press.
- Franchetto, Bruna. 2010. The ergativity effect in Kuikuro (Southern Carib, Brazil). In Spike Gildea & Francesc Queixalós (eds.), *Ergativity in Amazonia*, 121-158. Amsterdam: Benjamins.
- Gil, David. 2001. Escaping Eurocentrism: Fieldwork as a process of unlearning. In: Paul Newman & Martha Ratliff (eds.), *Linguistic Fieldwork*, 102-132. Cambridge: Cambridge University Press.
- Greenberg, Joseph H. 1963. Some universals of grammar with particular reference to the order of meaningful elements. In Joseph H. Greenberg (ed.), *Universals of Language*, 73-113. Cambridge, Mass.: MIT Press.
- Haider, Hubert. 1985a. The case of German. In Jindřich Toman (ed.), *Studies in German Grammar*, 65-102. Dordrecht: Foris.
- Haider, Hubert 1985b. A unified account of case and Θ -marking: the case of German.“ *Papiere zur Linguistik* 32. 3-36.
- Haider, Hubert. 1986. Who is afraid of typology? *Folia Linguistica* XX: 109-146.

- Haider, Hubert. 2000. The license to license. In Reuland, Eric (ed.), *Argument & Case: Explaining Burzio's Generalization*. 31-54. Amsterdam: Benjamins.
- Haider, Hubert. 2001. Heads and selection. In Norbert Corver & Henk van Riemsdijk (eds.), *Semi-lexical categories*, 67-96. Berlin: Mouton de Gruyter.
- Haider, Hubert. 2013. *Symmetry breaking in syntax*. Cambridge: Cambridge University Press
- Haider, Hubert. 2015a. "Intelligent design" of grammars – a result of cognitive evolution. In: Aria Adli & Marco García García & Göz Kaufmann (eds.): *Variation in language: System- and usage-based approaches*. 205-240. Berlin: de Gruyter.
- Haider, Hubert. 2015b. Head directionality – in syntax and morphology. In: Antonio Fábregas, Jaume Mateu, Mike Putnam eds. *Contemporary linguistic parameters*. 73-97. London: Bloomsbury Academic.
- Haider, Hubert. 2020. *Against severing the external argument from its verb*. [lingbuzz/005467]
- Haider, Hubert. 2021a. Grammar change – a case of Darwinian cognitive evolution. *Evolutionary Linguistic Theory* 3(1): 6-55.
- Haider, Hubert. 2021b. "OVS" – A misnomer for SVO languages with ergative alignment. lingbuzz/005680
- Haider, Hubert & Rositta Rindler-Schjerve. 1987. The parameter of auxiliary selection. *Linguistics* 25: 1029-1055.
- Haig, Geoffrey. 2008. *Alignment change in Iranian languages: A Construction Grammar approach*. Berlin: Mouton de Gruyter.
- Haig, Geoffrey. 2018. Northern Kurdish (Kurmanjî). In Haig, Geoffrey & Khan, Geoffrey (eds.) *The languages and linguistics of Western Asia. An areal perspective*, 106-158. Berlin: Mouton deGruyter.
- Hammarström, Harald. 2016. Linguistic diversity and language evolution. *Journal of Language Evolution*. 1. 19–29. doi:10.1093/jole/lzw002.
- Jackendoff, Ray. 1990. *Semantic structures*. Cambridge: The MIT Press.
- Janic, Katarzyna & Witzlack-Makarevich, Alena eds. 2021. *Antipassive. Typology, diachrony, and related constructions*. Amsterdam: Benjamins.
- Jones, Wendell & Paul Jones. 1991. *Barasano Syntax*. Studies in the languages of Colombia 2. Summer Institute of Linguistics & University of Texas at Arlington.
- Kalin, Laura. 2014. The syntax of OVS word order in Hixkaryana. *Natural Language & Linguistic Theory* 32, 1089-1104. DOI 10.1007/s11049-014-9244-x
- Keenan, Edward. 1976. Towards a universal definition of "Subject." In Charles Li (ed.), *Subject and topic*, 305-333. New York: Academic Press.
- Keenan, Edward L. & Bernard Comrie. 1977. Noun phrase accessibility and universal grammar. *Linguistic Inquiry* 8. 63-99.
- Keenan, Edward L. & Bernard Comrie. 1979. Data on the noun phrase accessibility hierarchy. *Language* 55. 333-351.
- Keenan, Edward L. & Cecile Manrohanta. 2004. Malagasy Clause Structure and Language Acquisition. [Proceedings of AFLA 11]. *ZAS Papers in Linguistics* 34. 178-202.
- Köhler, Oswin. 1981. La langue Kxoe. In Jean Perrot (ed.), *Les langues dans le monde ancien et moderne*, 483-555. Paris: Centre National de la Recherche Scientifique.
- Lahne, Antje. 2008. Excluding SVO in Ergative Languages: A new view on Mahajan's generalization. In Fabian Heck, Gereon Müller & Jochen Trommer (eds.), *Varieties of competition*, 65-80. *Linguistische Arbeitsberichte* 87. Leipzig: Universität Leipzig.
- Levin, Beth. 1983. *On the nature of ergativity*. Unpubl. MIT dissertation. Stanford: Center for the Study of Language and Information
- Madeira, Ana & Alexandra Fiéis. 2020. Inflected infinitives in Portuguese. In Ludovico Franco & Paolo Lorusso (eds.), *Linguistic variation: structure and interpretation*, 423-438. Berlin: Mouton De Gruyter.
- Mahajan, Anoop. 1997. Universal grammar and the typology of ergative languages. In Artemis Alexiadou & Tracy A. Hall (eds.), *Studies on universal grammar and typological variation*, 35-57. Amsterdam: Benjamins.

- Manning, Christopher D. 1996a. *Ergativity: Argument Structure and Grammatical Relations*. Stanford, CA: CSLI Publications/Cambridge University Press Dissertations in Linguistics series.
- Manning, Christopher D. 1996b. Argument structure as a locus for binding theory. *Proceedings of LFG-96, the First International LFG Colloquium and Workshops*. Grenoble, France.
- Marantz, Alec. 1981. *On the nature of grammatical relations*. Cambridge: MIT dissertation.
- Marantz, Alec. 1984. *On the nature of grammatical relations*. Cambridge: MIT Press.
- Marantz, Alec. 1991. Case and licensing. In Germán F. Westphal & Benjamin Ao & He-Rahk Chae (eds.), 234-253. *Proceedings of the 8th Eastern States Conference on Linguistics*. Ithaca, NY: CLC Publications.
- Martins, Silvana and Valteir Martins. 1999. Makú. In Robert M. W. Dixon, & Alexandra Y. Aikhenvald (eds.), *Amazonian Languages*, 251-267. Cambridge: Cambridge University Press.
- Mel'čuk, Igor, A. 2014. Syntactic Subject: Syntactic relations, once again. In Vladimir Plungjan, Mixail Daniël', Ekaterina Ljutikova, Sergej Tatevosov & Olga Fedorova (eds), *Jazyk. Konstanty, Peremennye* [Language. Constants, variables], 169-216. Sankt-Petersburg: Aleteja.
- Mithun, Marianne. 1991. Active/agentive case marking and its motivations. *Language* 67. 510-546.
- Mithun, Marianne. 2005. Ergativity and language contact on the Oregon Coast: Alsea, Siuslaw and Coos. In Andrew K. Simpson (ed.), *Proceedings of the Twenty-Sixth Meeting of the Berkeley Linguistics Society*, 77-95. Berkeley: Berkeley Linguistics Society.
- Mithun, Marianne. 2008. Does passivization require a subject category? In Greville Corbett & Michael Noonan (eds.), *Case and Grammatical Relations*, 211-240. Oxford University Press.
- Morse, Nancy L. and Maxwell, Michael B. 1999. *Gramática del cubeo*. Santafé de Bogotá: Instituto Lingüístico de Verano.
- Muysken, Pieter, Harald Hammarström, Olga Krasnoukhova, Neele Müller, Joshua Birchall, Simon van de Kerke, Loretta O'Connor, Swintha Danielsen, Rik van Gijn & George Saad. 2016. *South American Indigenous Language Structures (SAILS) Online*. Jena: Max Planck Institute for the Science of Human History. (<https://sails.clld.org>).
- Olawsky, Knut J. 2006. *A Grammar of Urarina*. Berlin: Mouton de Gruyter.
- Olawsky, Knut J. 2007. ObVioUS OVS in Urarina syntax. In: Andrew Simpson & Peter K. Austin (eds.), *Endangered languages*. [Special issue] *Linguistische Berichte*, 45-72. Hamburg: Buske Verlag.
- Paul, Ileana. 1999. *Malagasy clause structure*. Montreal: McGill University. PhD Diss.
- Pearson, Matthew. 2001. The clause structure of Malagasy: A Minimalist approach. UCLA Dissertations in Linguistics. (#21). Los Angeles: UCLA.
- Polinsky, Maria, Carlos Gómez Gallo, Peter Graff & Ekaterina Kravtchenko. 2012. Subject preference and ergativity. *Lingua* 122. 267-277.
- Polinsky, Maria. 2013. Antipassive Constructions. In Matthew Dryer, S. & Martin Haspelmath (eds.) *The World Atlas of Language Structures Online*. Leipzig: Max Planck Institute for Evolutionary Anthropology. (Available online at <http://wals.info/chapter/108>, Accessed on 2021-01-10).
- Polinsky, Maria. 2017a. Antipassive. In Jessica Coon, Diane Massam, & Lisa de Mena Travis (eds), *The Oxford handbook of ergativity*, 308-31. Oxford: Oxford University Press.
- Polinsky, Maria. 2017b. Syntactic ergativity. In Martin Everaert & Henk C. van Riemsdijk (eds.), *The Wiley Blackwell companion to syntax*, 2nd edn., 1-37. Hoboken, NJ: Wiley-Blackwell.
- Popovich, Harold. 1986. The nominal reference systems of Maxakalí. In Ursula Wiesemann (ed.), *Pronominal System*, 351-358. Tübingen: Narr.
- Queixalós, Francesc & Spike Gildea. 2010. Manifestations of ergativity in Amazonia. In Spike Gildea & Francesc Queixalós (eds.), *Ergativity in Amazonia*, 1-25. Amsterdam: Benjamins.
- Queixalós, Francesc. 2010. Grammatical relations in Katukina-Kanamari. In Spike Gildea & Francesc Queixalós (eds.), *Ergativity in Amazonia*, 235-283. Amsterdam: Benjamins.
- Rill, Justin. 2017. The morphology and syntax of ergativity: a typological approach. Newark: University of Delaware PhD dissertation.

- Roberts, Ian. 2021. Smuggling, ergativity, and the final-over-final condition. In Adriana Belletti & Chris Collins (eds.), *Smuggling in Syntax*, 318-352. Oxford University Press.
- Rojas-Berscia, Luis Miguel. 2014. A Heritage Reference Grammar of Selk'nam. Nijmegen: Radboud University. Master thesis.
- Rumsey, Alan. 1982. An intra-sentence grammar of Ungarinjin. *Pacific Linguistics*. Series b, no. 86. Canberra: National University of Australia.
- Schmidt, Wilhelm 1902. Die sprachlichen Verhältnisse von Deutsch-Neuguinea. In: *Zeitschrift für afrikanische, ozeanische und ostasiatische Sprachen* 6:1-99.
- Siewierska, Anna. 1996. Word order type and alignment type. *STUF - Language Typology and Universals* 49: 149-176.
- Siewierska, Anna. 2013. Passive Constructions. In: Dryer, Matthew S. & Haspelmath, Martin (eds.) *WALS Online* (v. 2020.4)
- Taraldsen, Tarald. 2017. Remarks on the relation between case-alignment and constituent order. In Jessica Coon, Diane Massam, & Lisa D. Travis (eds.), *The Oxford Handbook of Ergativity*, 332-354. Oxford: Oxford Univ. Press.
- Thráinsson, Höskuldur, 2007. *The Syntax of Icelandic*. Cambridge: Cambridge University Press.
- Tóth, Ildikó 2000. Inflected Infinitives in Hungarian. PhD dissertation. Tilburg University
- Trask, Robert L. 1979. On the origins of ergativity. In Frans Plank (ed.), *Ergativity: toward a theory of grammatical relations*, 385-404. London: Academic Press.
- Truswell, Robert 2014. Binding theory. In Andrew Carnie, Yosuke Sato & Daniel Siddiqi (eds.), *Routledge Handbook of Syntax*, 214-238. Abingdon: Taylor & Francis.
- Tsunoda, Tasaku. 2011. *A Grammar of Warrongo*. Berlin: Mouton de Gruyter
- Uslar, Peter K.U. 1889. Аварский язык. [Avar Language]. *Jazykoznanie III*. Tbilisi: Upravlenie Kavkazskago.
- Van Valin Jr., Robert D. 2005. *Exploring the syntax-semantics interface*. Cambridge: Cambridge University Press.
- Van Valin, Robert D. Jr. 1992. A synopsis of role and reference grammar. In Robert D. Van Valin Jr. (e.), *Advances in Role and Reference Grammar*. 1-166. Amsterdam: John Benjamins.
- Williams, Edwin. 1987. Implicit arguments, the binding theory, and control. *Natural Language and Linguistic Theory* 5. 151-180.
- Woolford, Ellen. 2000. Ergative agreement systems. University of Maryland: *Working Papers in Linguistics* 10:157-191.
- Yu, K.M. 2021. Tonal marking of absolutive case in Samoan. *Natural Language and Linguistic Theory* 39. 291-365
- Zwart, Jan-Wouter & Charlotte Lindenberg. 2021. Rethinking alignment typology. In András Bárány, Theresa Biberauer, Jamie Douglas & Sten Vikner (eds.), *Inside syntax III: Syntactic architecture and its consequences*, 23-50. Berlin: Language Science Press